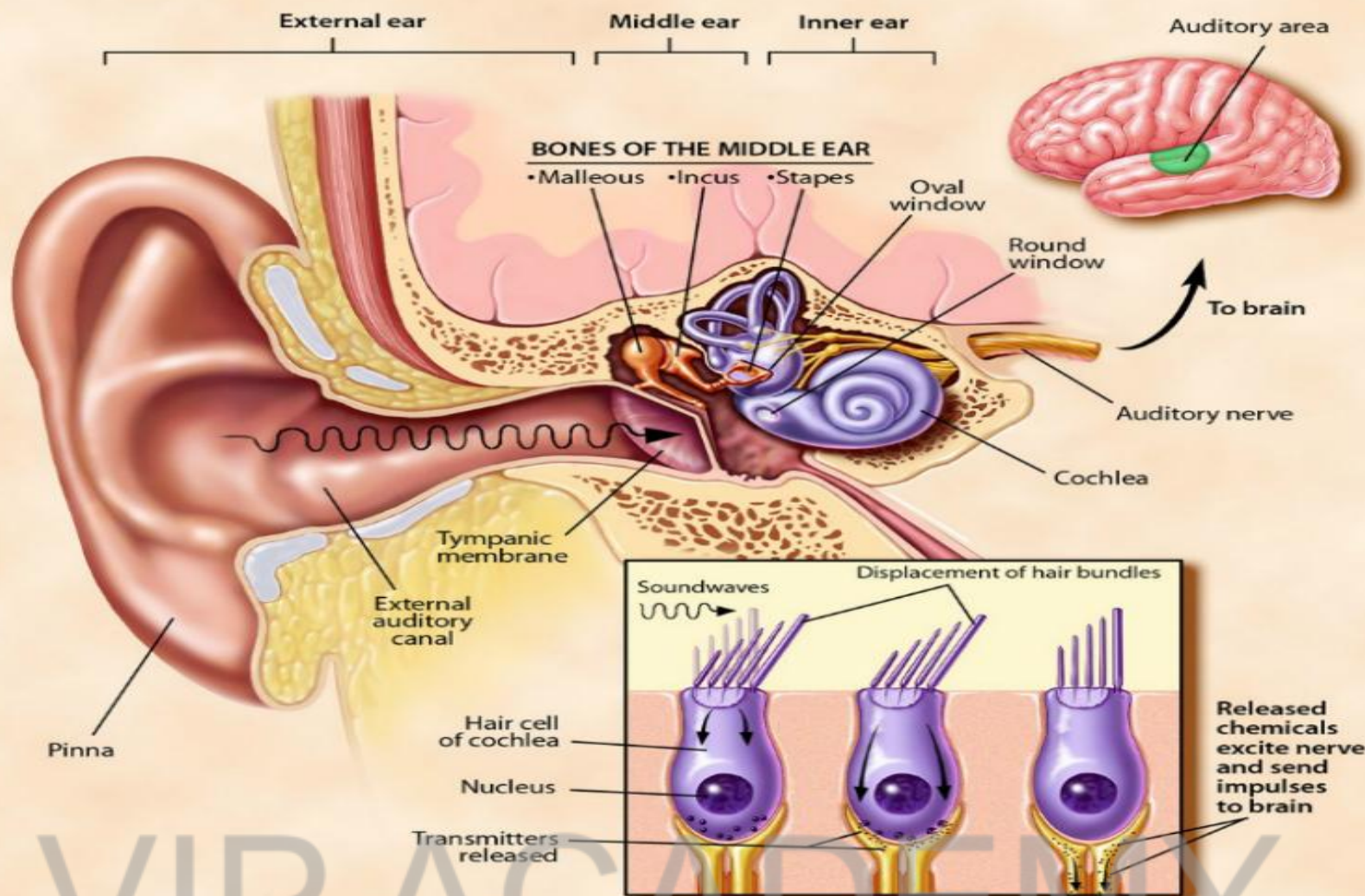
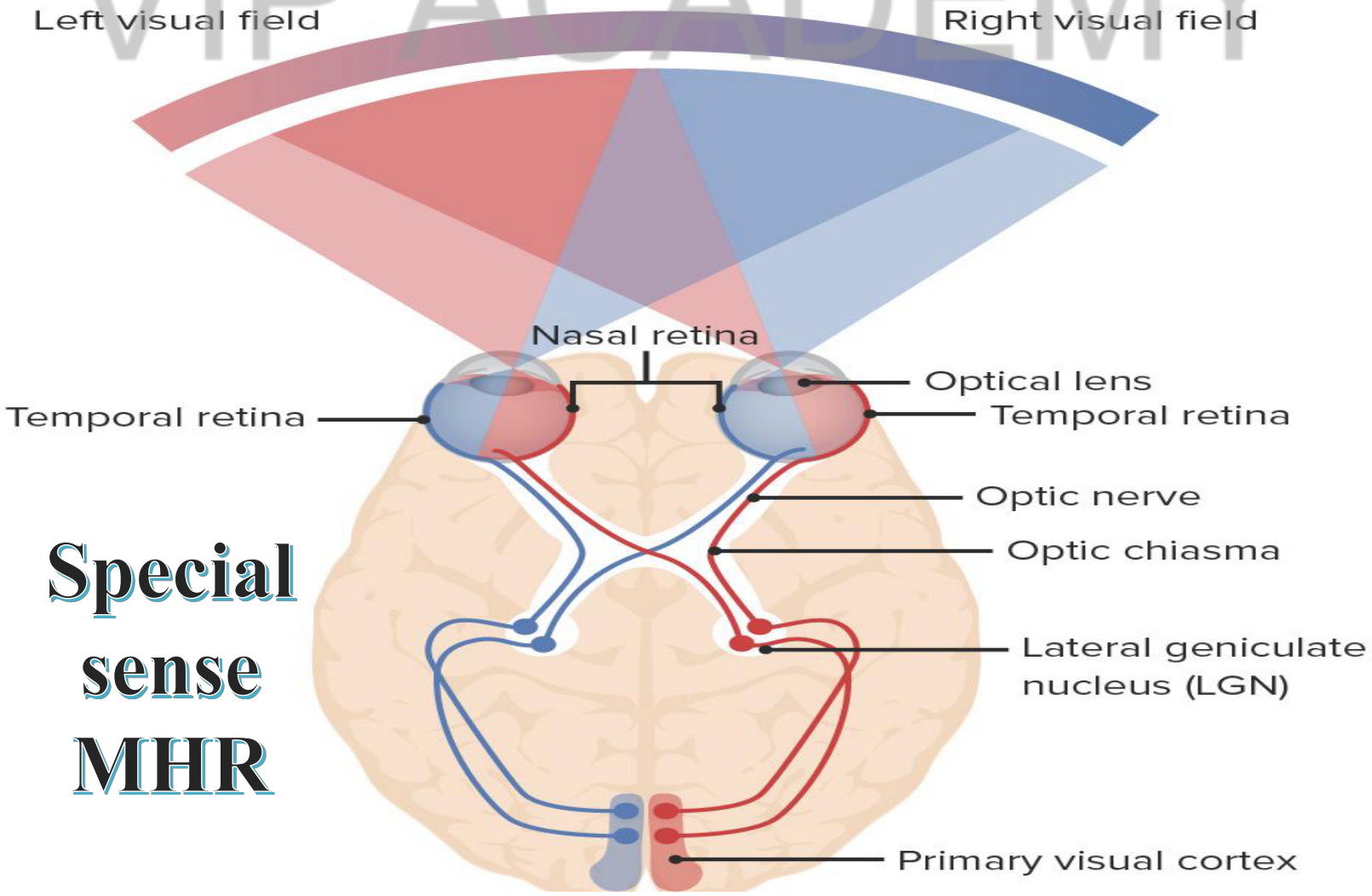




Left visual field

Right visual field



**Special
sense
MHR**

Special Senses

- Vision
- Hearing
- Smell
- Taste

Organs in the head

Protected by bone cavity

Information reach brain quickly

Visual axis

- The Line from fovea centralis to fixation point via the nodal point.

The axis along which the eye is directed

Light waves

- Visible spectrum of electromagnetic radiation: with wavelengths from 400-700nm
- **Spectral colors:** violet, indigo, blue, green, yellow, orange, red

Not perceived	Ultraviolet radiation	Cause • Darkening • Burning effect of skin • kills bacteria
	Infrared radiation	Cause heating effect on the body

- **Refractive index (RI):** ratio between velocity of light in air & **transparent** substance. RI of air= 1.0

- **Refractive power** (Degree of refraction) (bending) depends on

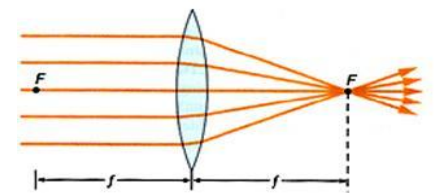
- ✓ Refractive index of the substance
- ✓ Angle of incidence (curvature)

No refraction for rays passing via nodal point

- ❖ **Focal point:** point behind the lens at which parallel rays converge in by convex Spherical Lens
diverging rays converge on point > focal length

- ❖ **Focal length**

- ✓ Distance between nodal point (central part of the lens) & focal point
- ✓ Depends on refractive power of the lens (inverse relation)
- ✓ **Refractive power**=reciprocal of focal length in meters (measured in diopter)



FLUID SYSTEM OF THE EYE (Aqueous, Vitreous humor)

- Maintains pressure in eyeball → to be distended.

A. Vitreous humor

Site Transparent gelatinous mass between lens & retina

Composition Contains albumin, hyaluronic acid (responsible for high viscosity).

(Substances diffuse slowly in vitreous humor, but there is little flow of fluid)

Functions

- ✓ Supports & Supplies the retina with glucose

B. Aqueous humor

Definition clear alkaline fluid → fills anterior & posterior chambers

Mechanism of Formation

➤ **Formed by** ciliary processes (projecting from ciliary body behind the iris)

➤ **Rate:** 2-3 $\mu\text{L}/\text{min}$ (continuously formed & reabsorbed)

1. **Active secretion:** $\text{Na}^+\text{-K}^+$ ATPase pumps Na^+ $\xrightarrow{\text{followed by}}$ passive diffusion of Cl^- , HCO_3^-
(formed by Carbonic anhydrase)
2. **Osmosis** of water: caused by Na^+ , Cl^- , HCO_3^-
3. **Diffusion** or active transport of glucose, amino acids

Composition differs from plasma (**less** glucose and protein) due to presence of blood aqueous barrier

Outflow

From posterior chamber *flow between ligaments of the lens* $\xrightarrow{\text{through pupil}}$ anterior chamber

→ iridocorneal filtration angle

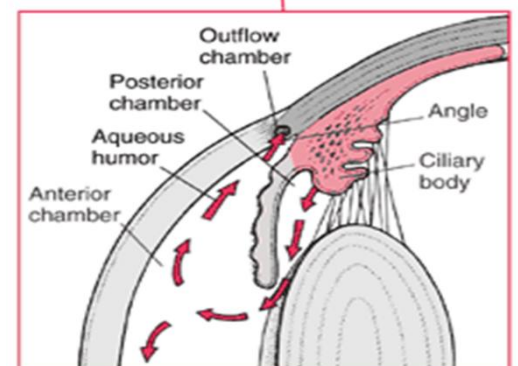
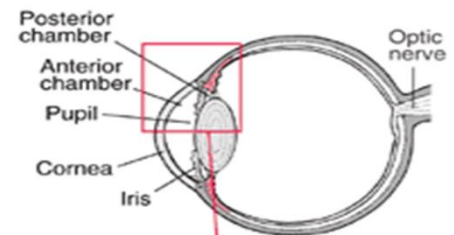
→ meshwork of trabeculae (spaces of Fontana)

→ canal of Schlemm

→ episcleral vein.

Functions

- 1) Supplies nutrients & O_2 to avascular cornea & lens.
- 2) Drains metabolites
- 3) One of the optical media of eye.
- 4) Maintains intraocular pressure.

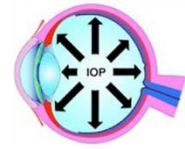


INTRAOCULAR PRESSURE

Definition relation between eyeball volume & ocular contents (Depend mainly on aqueous humor)

Normal value

- 12 – 20 mmHg.
- Diurnal variation
 - ✓ Highest in morning
 - ✓ Lowest in evening
- Maintained constant as **rate of drainage = rate of formation**



Regulation

1. Via variation in resistance to outflow from anterior chamber into canal of Schlemm
 - ↑formation → ↑IOP
 - open iridocorneal angle
 - distended spaces of Fontana
 - ↑drainage → ↓IOP
2. Pupil diameter e.g. pupil dilatation (mydriasis) → close the angle → ↑IOP (vice versa)

Functions

- IOP maintain
 1. Spheroid shape of eye
 2. Normal focusing mechanisms: → maintain lens flat during rest

↑IOP (glaucoma)	↓IOP
Stretch suspensory ligaments → no accommodation for near vision	→ relax suspensory ligaments → more spherical lens → more power

GLAUCOMA

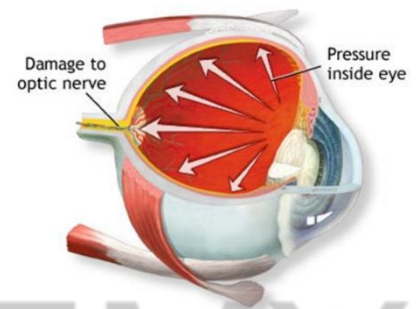
Definition ↑ intraocular pressure

Types & Causes

Open angle glaucoma	Close (narrow) angle glaucoma
↑ resistance to outflow via trabecular spaces into canal of Schlemm = Obstruction of drainage	Forward movement of iris

Results

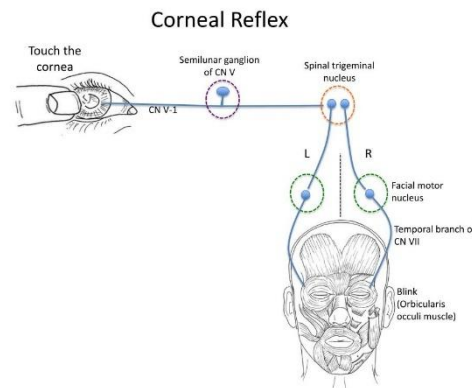
1. Disturbance of accommodation (focusing mechanism) for near vision
2. Severe pain.
3. Blindness (major cause) due to pressure on optic nerve & arteries



Refractive media of the eye Resting eye has an optical power of 67 D

Importance of cornea

1. Acts as fixed powerful convex lens **45 D** (2/3 of the refractive power), **due to**
 - High degree of curvature.
 - Refractive index = 1.38.
2. Regular curvature → helps formation of sharp retinal images
3. **Protective function:**
 - **Cornea reflex:** superficial reflex.
 - ✓ **Stimulus:** Touching cornea of one eye by foreign body
 - ✓ **Afferent:** ophthalmic division of trigeminal
 - ✓ **Efferent:** bilateral facial
 - ✓ **Response:** blinking of both eyes
 - ✓ **Importance:** Test integrity of trigeminal and facial nerves.
 - Absorb ultraviolet rays.



The Lens

Structure

- a. Biconvex: Posterior surface is more convex than anterior surface during rest.
- b. Formed of concentric layers of fibers
 - ✓ younger fibers → form lens cortex
 - ✓ old fibers → form lens nucleus
- c. Composition
 - ✓ Water = 60-70%
 - ✓ Proteins = 35%

Refractive power: 1/3 of the total refractive power during rest + high refractive index = 1.42+ not fixed power??

Metabolism

- Low metabolic rate
 - ✓ Energy from anaerobic glycolysis
 - ✓ Some aerobic metabolism occurs via presence of glutathione

Cataract



Definition partial or complete opacity (Loss of lens transparency)

Common causes:

- ✓ Old age (senile cataract)
- ✓ DM
- ✓ Excessive exposure to ultraviolet rays or temperature.

Treatment

- Surgical removal
- Replacement by powerful convex lens

Accommodation

Definition

- ✓ ↑ dioptric power via ↑ lens curvature
→ to focus image of near object on retina.

Mechanism

During rest:

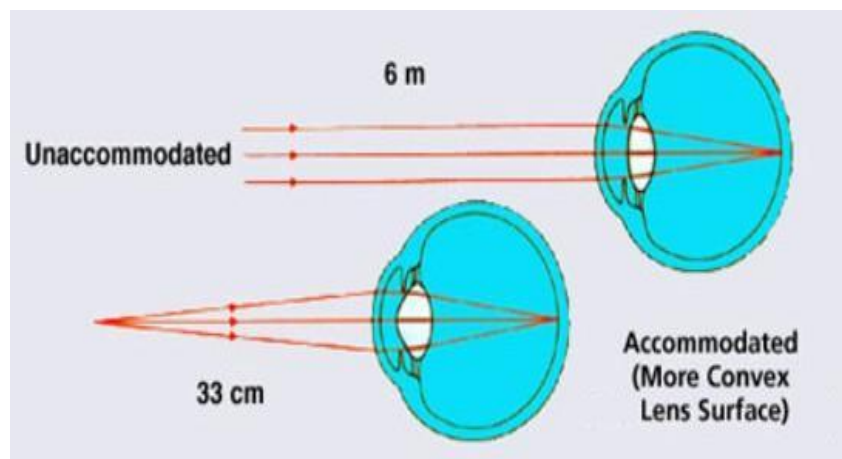
- ✓ The Lens has an elastic capsule → tend to be spherical
- ✓ Tendency is opposed by tension of suspensory ligament inside the muscular ring of the ciliary body

During near vision: < 6 meters

- ✓ Ciliary muscle contracts

Radial muscle fibers	Circular fibers
Extend from corneo-scleral junction to behind the insertions of lens ligaments in choroids When contract pull peripheral insertion of lens ligaments forward → ↓ tension on the lens.	Surround the lens as a ring When contract → ↓ circle → ↓ pull of the ligament = lax → ↓ tension on the lens → more spherical lens

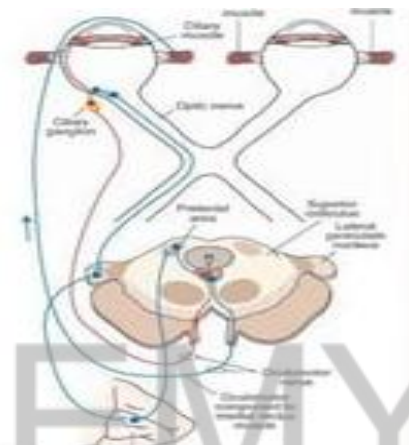
- ✓ Changes during accommodation, affects mainly anterior surface



Near Response (Near Reflex)

1. Accommodation	↑dioptric power of the lens.....
2. Convergence	Contraction of medial recti muscles of both eyes to focus images of near object on fovea centralis of each eye. (gaze of 2 eyes shift towards the object)
3. Miosis	Contraction of 2 constrictor pupillae muscles

• Pathway of near response

Receptors	Photoreceptors (rods & cones)
Afferent	Optic nerve → optic chiasma (crossing of nasal fibers) → optic tract → relays in lateral geniculate body of thalamus. → optic radiation → visual sensory area (17) → visual association area (18, 19) → then to 
Center	superior colliculus in midbrain
Efferent	From superior colliculus to both Edinger Westphal nucleus (EDW) → preganglionic parasympathetic in oculomotor nerves → ciliary ganglion → postganglionic short ciliary nerves → contraction of ✓ Ciliary muscle (accommodation) and ✓ pupillo-constrictor muscle (miosis) ○ Motor fibers of oculomotor nerve → contraction of medial recti → convergence ○ Parasympatholytic drug (atropine): Causes ✓ Paralysis of accommodation (cycloplegia) & ✓ Pupillary dilation

Power of Accommodation

Difference between

- Lens power during maximal accommodation (near vision) &
- Lens power during rest (far vision)

↓with age due to ↓elasticity of the lens			
At 10 years	20 years	50 years	60 years
14D	10 D	2D	1D

Near Point	Far Point
Nearest point seen clearly with maximum accommodation= 10 cm in young adults 80 cm at 60 years (↑with age)	First point seen clearly without accommodation= 6 m or more in emmetropes.
Range of Accommodation= distance between far and near points (↓with age)	

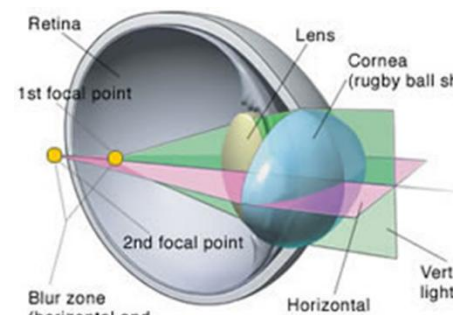
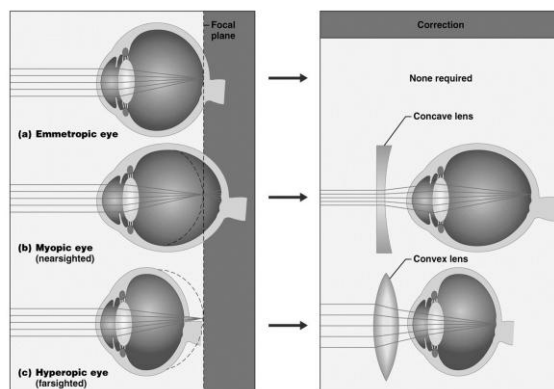
Presbyopia : see later

Errors of refraction

📌 **Emmetropic eye** = normal eye

- Distant objects: seen clearly when ciliary muscle is relaxed
- Near objects are seen clearly when ciliary muscle is contracted

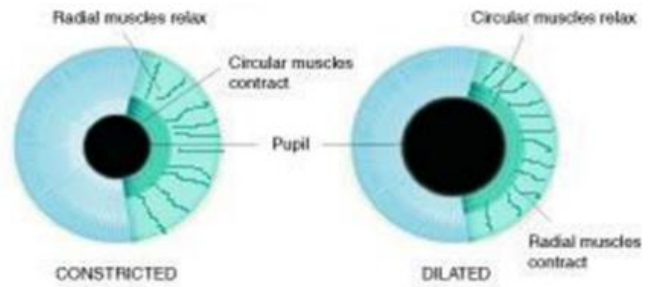
Error	Myopia Near sightedness	Hypermetropia Far sightedness	Presbyopia	Astigmatism
Definition:	Parallel rays focus in front of retina	Parallel rays focus behind the retina	↓ power of accommodation by age	
Cause	Great refractive power of the eye (↑ curvature of lens & cornea) Long eye ball	Weak refractive power of the eye (mainly cornea) Short eye ball Accommodation reflex work all the time	↓ elasticity of lens due to protein denaturation	Inequality of <u>corneal</u> curvature in various planes (egg shaped)
Optical changes			Near point: > 10 cm more distant Far point: normal (no change)	Refractive Power is different in different axes
Correction			Convex lens for near vision only	



Uveal Tract

✚ Middle layer, composed of:

- Iris,
- Ciliary body
- Choroid



Iris

✚ **Pigmented** disc contains 2 involuntary muscles:

	Constrictor pupillae muscle	Dilator pupillae muscle
Fiber	Circular	Radial
contraction→	Miosis	Mydriasis
Supplied by	Parasympathetic fibers (oculomotor nerve)	Sympathetic fibers (from upper 2 thoracic segments).

Functions of the iris

1- **Iris pigment restricts light passage** to the pupil

Regulate amount of light entry via pupil: from

1.5 mm during pupil constriction to 8 mm during dilatation.

2- **Protection:** superficial light reflex→ protect retina

3- **Prevents spherical & chromatic aberrations:** ↓light rays falling on peripheral parts of lens

Spherical aberration	Chromatic aberration
• lens curvature at periphery > center • lens Power at periphery > center. • light rays pass through periphery focus closer on different point →blurred image	• Peripheral parts of lens acts as a prism (analysis white light into spectral components (each have different focus) ✓ Violet rays refract more than red rays. ✓ Image is surrounded by halos.

4- **Constriction of pupil ↑depth of focus =**

- Maximum distance along an object can move while lens power is constant
- Small pupil →cause all light rays to pass via center of the lens.
Central rays are always in focus.

Pupillary light reflex

Receptors	Photoreceptors
Afferent	optic nerve → optic chiasma → optic tract (No relay in lateral geniculate body)
Center	pretectal nucleus of midbrain
Efferent	Pretectal nucleus → EWN of both sides → Preganglionic fibers in oculomotor nerves → ciliary ganglion → postganglionic ciliary nerves → contraction of both constrictor pupillae muscles
Response	Direct pupillary light reflex: constriction of the stimulated eye Indirect or consensual light reflex: constriction of unstimulated eye, due to: • Partial crossing at optic chiasma • bilateral innervation of EWN from pretectal nucleus

Argyll-Robertson pupil

Cause neurosyphilis due to **lesion in pretectal nucleus**.

Manifestations

- ✓ Pupillary constriction on near reflex (preserved accommodation reflex) +
- ✓ **No** Pupillary constriction on light reflex (loss of light reflex)

Causes of	Miosis	Mydriasis
Physiological	Light reflex Near (accommodation) reflexes. Parasympathetic predominance (Sleep)	Dark adaptation Distant (far) vision Sympathetic (fear, emotions, asphyxia, pain)
Drugs	Parasympathomimetic: pilocarpine, eserine Morphine → marked constriction Histamine → direct action on constrictor	Sympathomimetic: act on α adrenergic receptor in dilator pupillae Para sympatholytic: (atropine) → (Block muscarinic receptors in constrictor pupillae) Alcohol, chloroform poisoning
Neurological	Lesion of cervical sympathetic ganglion (Horner's syndrome)	Lesion of oculomotor nerve or EDW
Anesthesia	3 rd stage	2 nd & 4 th stage

Changes of pupil size during anesthesia

	pupil size
First stage	normal & react to light
Second stage	dilated due to ↑ sympathetic activity
Third stage	Constricted (surgical stage).
Fourth stage	Dilated again- loss of light & corneal reflex. (death may occur)

[B] Choroid: Function

- 1- Vascular layer** → provides blood supply to the eye.
- 2- Rich in melanin**
 - forms dark coat inside eyeball
 - prevents reflection of light rays inside the eye.
- 3- Gives attachment to ciliary muscles.**

(C) Retina

- Light sensitive part of the eye
- Between choroid and vitreous
- Contains visual receptors.

The pigment layer of the retina

- **In direct contact with the choroid**
 1. Prevent light reflection inside eye ball (not absorbed by rods & cones)
→ prevent blurring of vision
 2. Store large amount of vitamin A (precursor of photosensitive pigments)
 3. Phagocytose membranous disc & debris sloughed from photoreceptors

Horizontal cells	Amacrine cells
Stimulated by rods and cones Axons synapse with dendrites of bipolar cells & terminals of rods and cones located laterally	Stimulated by bipolar cells & Excite ganglion cells.
Inhibitory cells → secrete inhibitory transmitter (GABA)	Excitatory cells → secrete excitatory transmitter.
When stimulated → inhibit rods, cones and bipolar cells lateral to the stimulated ones (lateral inhibition) to sharpen the image	Respond very strongly & adapt very rapidly. Important in detection of ✓ Sudden changes in light intensity and ✓ Moving objects.

N.B.:

1. Retinal vessels supply bipolar & ganglion cells,
but photoreceptors via capillary plexus in choroid.
 - Retinal detachment → is damaging to the receptors.
2. Arteries, **arterioles** and veins in superficial layer of retina near vitreous can be seen via ophthalmoscope.
 - **The only place** where arterioles can be seen.
 - Ophthalmoscopic examination → diagnose vascular disorders as in **DM and hypertension**.

Parts of Retina of Physiological Importance

A. Macula Lutea

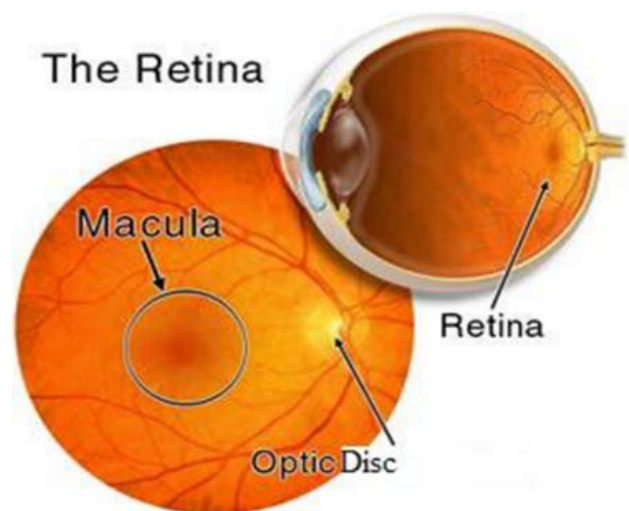
- ✓ yellow spot $< 1 \text{ mm}^2$
- ✓ 3 mm temporal to optic disc, opposite to posterior pole of the eye.

B. Fovea centralis

- ✓ Central part of macula (0.4 mm) →
Contain densely packed cones only
- ✓ Surrounding the fovea
 - Rods gradually appear & predominate
 - Cones decrease
- ✓ High visual acuity due to:
 1. Light falls direct on cones as
 - all layers are displaced (the retina is thinned out)
 - cones are not covered by blood vessels.
 2. Absence of convergence
Ratio of cone to bipolar to ganglion cell = 1:1:1 → one optic nerve fiber to the brain
 3. Pigmented layer is Highly developed → markedly ↓ light reflection

C. Optic Disc

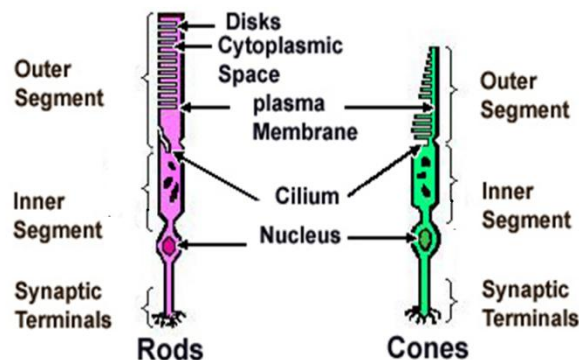
- ✓ Exit of optic nerve (1.5 mm)
- ✓ 3mm nasal (medial) to macula
- ✓ Contains no rods and cones
→ cause physiological **blind spot**



Characteristics of Rods and Cones (Photoreceptors)

	Rods	Cones
Sensitivity Threshold	Highly Sensitive to low light intensity low threshold of stimulation (stimulated by dim light) as: a. Rods contain more pigments → absorb more light than cones → produce greater response b. Rods have longer receptor potential than cones; → allow summation of response easily	Less sensitive to light high threshold (stimulated by bright light) as: a. Cones contain less pigments b. have shorter receptor potential
Responsible for	dim (night) vision or scotopic vision	bright (day) or photopic vision.
convergence	Many rods (200) converge on single bipolar cell and many bipolar cells converge to one ganglion	Each cone in fovea centralis → one bipolar → one ganglion cell
visual acuity	Low	High
Details	Can't perceive details of objects	Details of objects are perceived
Colors	Can't perceive colors	Colors can be perceived
Duplicity theory of vision: Von Kries theory?	<u>Scotopic vision</u> dim light vision. Function of rods. Details, boundaries and colors can't be detected.	<u>Photopic vision</u> : day vision. function of cones. Details, boundaries and colors can be detected

- Light from a star cannot be seen if fixed on fovea
- but becomes visible if vision is directed to bring star image on the rods.



Spectral Sensitivity of Photopigments

Purkinje Shift Phenomenon:

Definition:

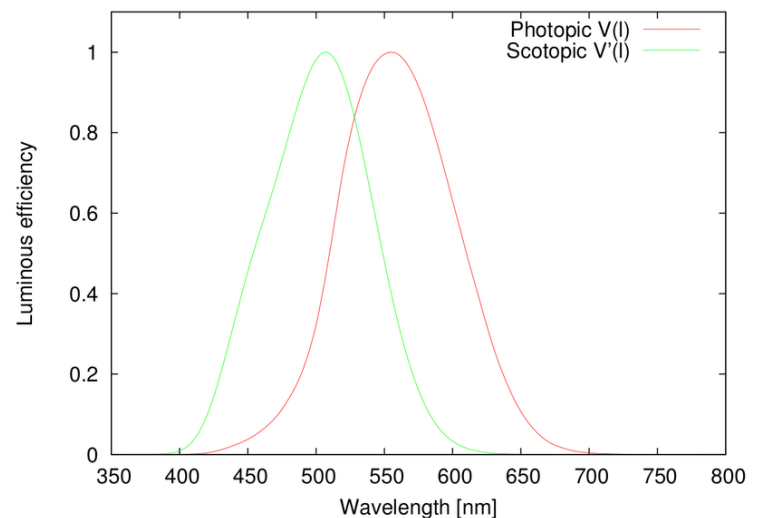
Shift of spectral sensitivity

- From maximal sensitivity of scotopic function at 505nm
- To maximum sensitivity of photopic function at 550nm.

studied by the luminosity curves

During photopic vision	During scotopic vision
Different wavelengths of the spectrum are perceived as different colors	All wavelengths appear as shade of grey. Red is not seen at all.
Maximal retinal sensitivity is obtained At wavelength 550nm At yellow green zone of spectrum.	Maximal retinal sensitivity is obtained At wavelength 505nm At blue green zone of spectrum
yellow part appears most luminous	green blue part appears most luminous.
characteristic of absorption spectra of 3 cone pigments	characteristic of absorption spectrum of rhodopsin.

- During daytime: Red & blue flowers are almost equally bright
- but as light fades
 - ✓ blue flowers are seen clearly
 - ✓ red flowers appear black.



Photoreceptor Potentials & Signal Transmission in Retina

- a- Photochemical changes (Bleaching).
- b- Genesis of electrical response & Excitation cascade.
- c- Signal transmission in the retina.

A. Photochemical changes (Bleaching)

Photopigments

Present in outer segment of photoreceptor (One type **in rods**), (3 types **in cones**)

Consists of 2 parts:

Opsin	Chromophore
	Carotenoid pigment (11-cis retinal); Derivative of vitamin A → has <u>curved configuration</u> light-sensitive part of photopigment.
Different in each type	The same in all types

Changes

Rhodopsin $\xrightarrow{\text{light}}$ batho-rhodopsin

→ lumirhodopsin

→ metarhodopsin I

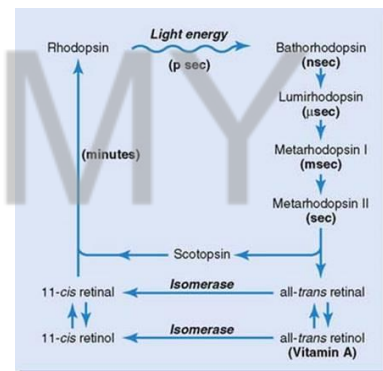
→ metarhodopsin II (active rhodopsin) (initiate electrical change)

→ separation of all trans-retinal (straight configuration) from Opsin (Bleaching)

- (independent on light): All trans-retinal $\xrightarrow{\text{retinal isomerase}}$ 11cis-retinal
- 11cis-retinal + scotopsin **Rhodopsin**

Role of Vitamin A in Rhodopsin Formation

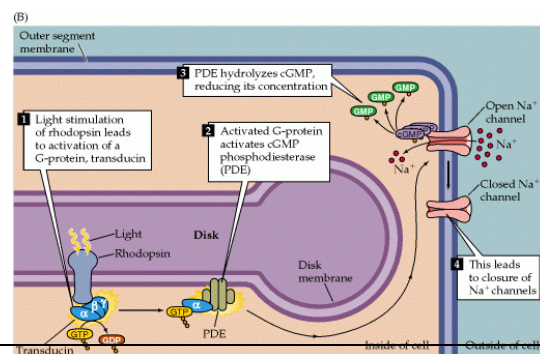
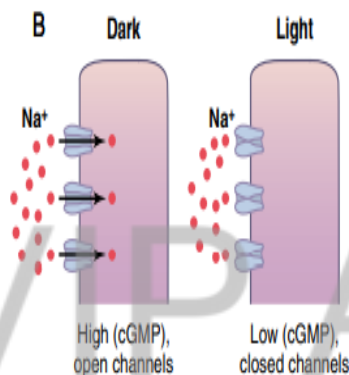
- a. **Vitamin A** present in **rods & pigment** layer → supply new retinal to form **more rhodopsin**
- b. Rods are more sensitive than cones to Vitamin A deficiency (↓ formation of retinal & rhodopsin)
→ leads to night blinding (nyctalopia) as rhodopsin is essential for night vision.
Untreated vitamin A deficiency end in day and night blindness.



B-Electrical Response in Retina

Dark current

- Inner segments of receptor
 - pump Na out
- Outer segments contain Na⁺ channels (maintained opened by cGMP)
 - cause continual leak back of Na⁺
 - ↓ negativity to -40mV
 - depolarization
 - continuous release of transmitter (G)



C. Excitation cascade (Transduction process)

On light exposure

- **Metarhodopsin II** (activated rhodopsin)
 - activates G-protein (transducin) in cell membrane of rods & discs
 - activates phosphodiesterase
 - hydrolyses c-GMP to 5-GMP
 - ↓ c-GMP
 - closure of Na⁺ channels
 - ↓ Na⁺ conductance into outer segments
- Na⁺ continue to be pumped out of inner segment
 - hyperpolarization (-70mV at maximal light intensity)
 - ↓ release of **glutamate**

(that act as excitatory or inhibitory depending on the channels on bipolar cell)
- Degree of hyperpolarization α log intensity of light
 - to discriminate wide range of light intensities.

- Receptor **potential is a state of hyperpolarization** (not depolarization)

On removal of light

- Inactivation of metarhodopsin II by rhodopsin kinase
 - inhibits phosphodiesterase & resynthesis of c-GMP by guanylate cyclase → Na⁺ channels reopen → membrane potential -40mV

D. Signal Transmission in the Retina

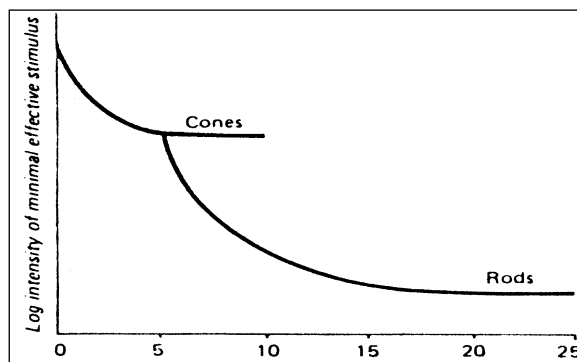
- Photoreceptors electrical response is **local graded potential (generator potential)**, (not action potentials)
- Local potential is passed from one neuron to other neurons (horizontal, bipolar and most amacrine cells)
 - then to ganglion cells
- **Only ganglion cells and some amacrine cells, conduct signals as action potentials.**
- *Each ganglion cells receive input from many bipolar cells which inturn receive inputs from many receptors = organized into circular receptive field*

Automatic Regulation of Retinal Sensitivity (retinal adaptation)

Definition ability of retina to adjust sensitivity to different light intensities (2 types)

	Light adaptation	Dark adaptation (max in 30 mins)
Cause	• Suddenly Shift from dim to bright light • At first: light → intensely bright and uncomfortable. • Within 5 minutes → ↓ photopigments → ↓ retinal sensitivity, ↑ retinal threshold → acute vision returns	Shift from bright to dim light at first: nothing is seen , then outline of objects is gradually identified Reach maximum in 30 minutes (vice versa)
Changes occurring	1. Miosis 2. Breakdown of photopigment 3. ↓ sensitivity of rods & cones to light 4. ↓ signal intensity in retinal neurons.	1. Mydriasis 2. Regeneration of pigment 3. ↑ retinal sensitivity of rods & cones 4. ↑ signal intensity in retinal neurons
		2 Components: • Rapid small rise of retinal sensitivity (during 1st 5-10 min.), caused by dark adaptation of cones • Slower greater rise in retinal sensitivity (30 minutes), caused by dark adaptation of rods (full loading of pigment)

- Retina ↑ its sensitivity during maximal dark adaptation (low light intensity) by 100,000–500,000 times its sensitivity during light adaptation (high light intensity).



COLOR VISION

Definition ability to discriminate between different wave lengths (colors)

Function of the cones

Primary color red, blue, green

- Mixing with different proportion → produce white or any spectral or extra-spectral colors

Complementary color pairs of colors when mixed → produce white e.g. yellow is complementary for blue

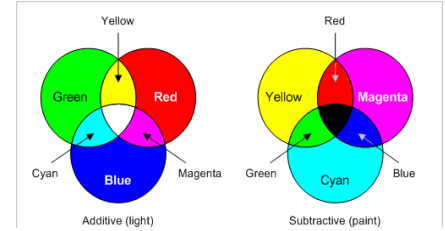
For any color, there is a complementary color

Color mixing is mixing wave length (not mixing paints or pigments)

Normal eye discriminate 7 color of spectrum & 100 intermediate color

White color: produced when 7 spectral colors in same proportion as in sunlight.

Black: produced by absence of light



Mechanism of color vision: (Young-Helmholtz theory) or **trichromatic theory of color vision**

- 3 types of cones.
- Each contains different pigment (**3 cone pigments**): (photopsin + retinal)
→ maximally sensitive to only one of 3 primary colors.

Blue sensitive pigment (Short wave pigment):	Green sensitive pigment (Middle wave pigment)	Red sensitive pigment (Long wave pigment):
Absorbs light maximally in blue violet portion (wavelength 445nm).	absorbs light maximally in green portion (Wavelength 535nm)	Absorbs light maximally in yellow portion red portion (wavelength 575nm).

Principle of cones stimulation by different colors

- Colored light stimulates 3 cones unequally depending on wavelength
- Relative frequency of impulses from 3 cones reaches visual cortex → sensation of color e.g.

Orange light (wavelength 580nm), Stimulates

Red cones by 99%,

Green by 42%

Blue by 0%.

Ratio of stimulation (99: 42: 0) → orange

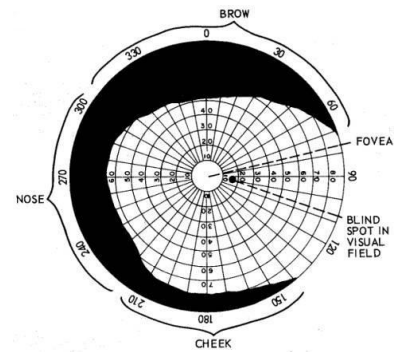
- Equal stimulation of all types of cones → white color.
- Color perception starts in retina, Color coding depends on relative activity in each of 3 cones

Neural pathway of vision (Visual Pathway)

Photoreceptors	→ bipolar cells → ganglionic cells (axons form optic nerve) - Nasal retina receives rays from temporal visual field (vice versa) - Upper half of the retina receives light rays from lower visual field (vice versa)
Optic nerve	→ Optic chiasma (nasal fibers cross / temporal fibers on ipsilateral side)
Optic tract	Contains fibers from ipsilateral temporal retina + contralateral nasal retina Carries impulses from opposite same halves of visual fields of both eyes. optic tract fibers terminate in: - Suprachiasmatic nucleus of hypothalamus →controlling light evoked endocrine functions (circadian rhythm). - Pretectal nucleus →for pupillary light reflex. - Superior colliculus →for reflex eye movements. - Lateral geniculate body
LGB → optic radiation → visual cortex in occipital lobe:	
Primary visual cortex (area 17, striate cortex)	decodes information about contrast, color, depth, form and movements - Right visual cortex receives from right halves of both retina i.e. temporal of same side & nasal of opposite side. (field?????) - Macula: occupies the posterior & occipital pole & widely represented
Visual association areas (2 ^{ry} visual area 18 & 19, peristriate cortex):	They are responsible for: - Detection of the nature of the object (e.g. ball or pencil). - Visual orientation and depth perception i.e. localization of objects in space. - Sending information to other areas of cerebral cortex.

Visual Field

- Area of external world can be seen by an eye without moving it.
- Extends
 - ✓ 50 degrees upwards
 - ✓ 80 degrees downwards
 - ✓ 60 degrees nasally
 - ✓ 90 degrees temporally



Binocular Vision

Definition seeing one object with 2 eyes at same time without diplopia (double vision).

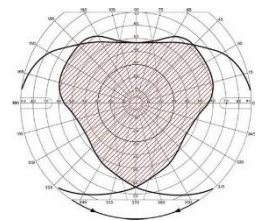
- **Monocular field of vision:** is a crescentic area 35 degrees at outer part of each temporal field.
- Light rays from objects in these regions fall on the nasal side of the retina of the near eye.

Factors affecting binocular vision

- 1- Overlap of central parts of both visual fields.
- 2- nearly equal dioptric power of both eyes
- 3- Nasal part of the fovea in one eye is corresponding to temporal part of fovea in other eyes
- 4- Normal extraocular muscles to move eye balls → so Images fall on corresponding points of 2 retina
- 5- Normal visual cortex for fusion of 2 images

Advantages

- 1- Wider visual field than monocular vision
- 2- Minimize defect in retina whether physiological or pathological
- 3- Abnormal refraction of one eye is masked by the normal eye.
- 4- Better depth perception and stereoscopic vision.



Depth perception

Definition determination of distance by visual system

(Monocular property, more accurate by binocular vision),

Depends on

- 1- **Size** of objects: Change with distances
- 2- **Color** and details: Fade with distances
- 3- **Occlusion** of part of distant object by nearer one.
- 4- Distribution of **light** and shade
- 5- **Shadows** which an object casts upon its surroundings.
- 6- **Moving parallax** (when head move in certain direction)



Nearer objects move in opposite direction

Distant objects move in same direction

- 7- **Parallel lines converge with distance.**

Stereoscopic Vision

Definition differentiation of 3 dimensions of objects (depth, height and width),

Perceived by monocular vision (better by binocular vision)

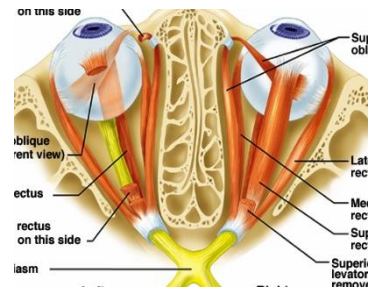
Cause corresponding points in 2 retinas receives slightly dissimilar images from any object

as 2 eyes are few centimeters away from each other

→ (2 Slightly dissimilar images fuse in cortex) → give stereoscopic vision

Depends on

- 1- Factors that determine depth perception ????
- 2- Formation of slightly dissimilar images on both retinas



Strabismus or Squint

Cause incoordination of external ocular muscles → images **not fall on corresponding points** → diplopia.

Complications

- If occurs chronically in children < 6 years,

image from squinting eye is suppressed by visual cortex → diplopia disappears (very low acuity in this Eye)

If not treated before 6 years → **permanent loss** of vision of this eye

Treatment Eye muscle training **exercises** / Glasses with special **prisms** / **Surgical** correction

HEARING (AUDITION)

 **Sound** is a stimulus for hearing

- ✓ Produced by vibrating object that cause alternating compression and rarefaction in the medium
- ✓ Represented in a wave with peaks and bottoms
- ✓ **Sound wave:** not cause any net flow of air in the direction of motion
air particles move to and fro around its main position

 **Velocity:** higher in solids > water > liquid and
zero in vacuum.

Characteristics of sound wave

a. **Wave length:** distance between 2 successive peaks or bottoms

b. **Frequency (pitch):** measured in Hertz (Hz) i.e. cycle/ sec.

Pitch correlates with frequency i.e. high frequency gives high pitched sound.

- Range of human hearing is 20- 20000 Hz.
- Range of the average speaking voice is 2000- 5000Hz. [2- 5 KHz].
- In other animals, as bats and dogs, can hear much higher frequencies

c. **Amplitude** (intensity) of sound: It has the subjective correlate of **loudness**

- Unit of intensity is decibel (dB)
- P sound= the pressure of the sound stimulus
- P_{spl}=sound pressure level at the threshold for human hearing

$$dB = 20 \log \frac{P_{sound}}{P_{spl}}$$

Decibel	index
0	Threshold of hearing (standard sound reference level)
40	Talking or whisper
60	Normal conversation
80	Traffic noise
100	Pneumatic drill
120	Jet plane
> 140	Painful damaging to auditory receptors

Functions of external ear

1. Collection of sound stimuli.

2. ↑sound pressure at tympanic membrane:

- ✓ external ear (trumpet) collects sound over large area
→ concentrates it in smaller area of external meatus

3. Sound localization: mainly by higher cortical interactions.

- ✓ But shape of pinna helps in determining source of sound.

4. Protective function:

A. Skin lining auditory canal secretes **wax**

- ✓ Has bactericidal effect
- ✓ Prevents entrance of insects

B. Air in external canal is kept moist, near body temperature for proper functions of tympanic membrane

5. Function of tympanic membrane:

- ✓ Thin cone-shaped membrane (1 cm)
- ✓ Transmits vibrations from outside to middle ear.

• Characters of the membrane:

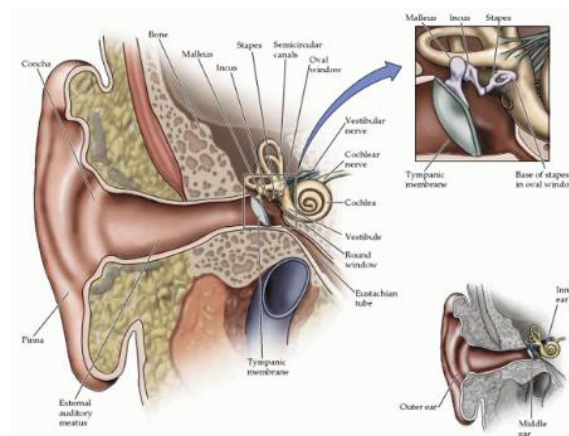
A. Elastic

B. Tense

C. Aperiodic [has no natural frequency]

rate of vibrations = rate of sound frequency (acts as resonator)

D. Damping = stops vibrations when the sound stops



Function of middle ear

1. Middle ear couples sound energy to cochlea

- ✓ Vibrating tympanic membrane → vibrate middle ear bones → vibrate cochlear fluid

2. Physical protection of cochlea

3. Impedance matching

- ✓ by amplifying sound pressure to match impedance of air to higher impedance of cochlear fluids

✓ Mechanism

A. Force collected over **large area of tympanic membrane** [55mm²] →

is concentrated over **smaller** area of footplate of stapes in **oval window** (3.2mm²)

→ ↑ pressure [**17 times**]

B. Ossicles act as a **lever** (malleus longer > incus)

→ minimum displacement of stapes → ↑ pressure [**1.3 fold**]

C. $17 \times 1.3 = \mathbf{22 \text{ times}}$ ↑ in pressure exerted on cochlear fluid.

4. Ossicular system apply sound pressure to only one window of cochlea

- ✓ Cause **differential pressure** between 2 windows
(required for **movement** of cochlear fluid)

5. Function of middle ear muscles:

A. Handle of malleus is constantly pulled inward by tensor tympani muscle

→ keeps **tympanic membrane tensed** → for optimum vibration

B. **Attenuation of sound (Acoustic reflex or tympanic reflex):**

- **Tensor tympani** innervated by cranial nerve V, **stapedius** by cranial nerve VII

- **Muscles** contract in response to intense sounds

→ pull malleus inwards & stapes away from oval window

→ dampen movement of ossicular chain & sound transmission.

➤ Significance

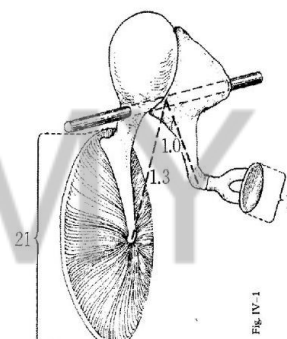
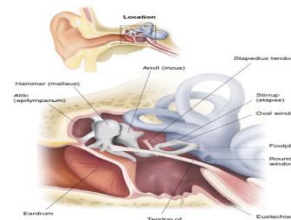
A- Protect auditory receptors from damage

- ✓ Reaction time for reflex = 40-160 ms.

- ✓ **Does not protect against brief intense sound e.g. as by gunshots.**

B- Muscles contract prior to vocalization and chewing

→ ↓ intensity of sounds produced by these activities.



6. Function of Eustachian tube connects middle ear with nasopharynx,

- ✓ Normally, closed at rest
- ✓ Opens during swallowing, chewing or yawing

A- Causes renewal of air in middle ear

B- Equalizes pressure on both sides of the drum.

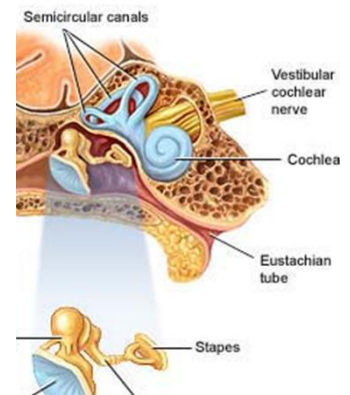
- if Eustachian tube is blocked during flying, diving or middle ear infection
 - cause Pressure difference
 - displacement, pain, ruptures of tympanic membrane

C- Lined by ciliated epithelium (act in one direction) + oblique direction of the tube (in adults).

→ help drainage of middle ear secretion

NB: In children, tube: short, wide, straight

→ easy entry of nasopharyngeal secretion & fluids → cause middle ear infection



Inner ear [labyrinth]

- ✓ Contains **vestibular** and **auditory** receptors (**cochlea**).

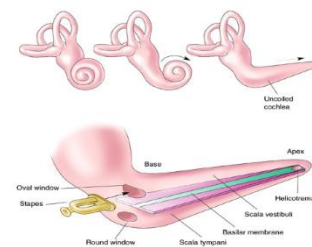
Cochlear organization

Perilymph: similar to ECF (high in Na^+ and low in K^+)

Endolymph: similar to ICF

Endo-cochlear (endolymphatic) potential

- ✓ Endolymph in scala media is +80 in comparison to perilymph
- ✓ Cause: due to continuous transport of K^+ into scala media by **stria vascularis** (covers lateral wall of scala media)



Organ of Corti: receptor of hearing

Site on top of basilar membrane from apex to base of cochlea

Contains short hair cells (**auditory receptor**)

1. **Tectorial membrane** is attached to modiolus at one end and rests on the hairs.

2. **Tight junctions** between hair cells & supporting cells

- ✓ Prevent endolymph from reaching bases of hair cells
- ✓ **Bases** of hair cells are bathed in perilymph
- ✓ **Processes** are bathed in endolymph

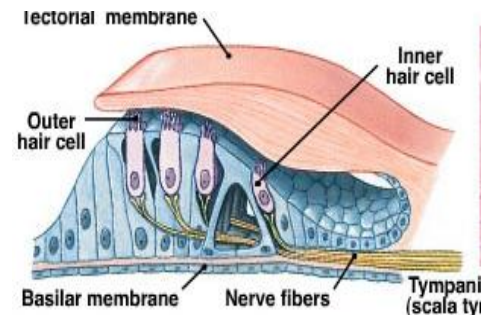
3. **Hair cells potential**

- ✓ = -70 mv with respect to perilymph
- ✓ = -150 mv with respect to endolymph at their tips
- ✓ Importance: → **sensitize** hair cells → ↑ their ability to respond to slightest sound.

4. **Cochlear nerve fibers [afferent & efferent]**

- ✓ Enter via tunnel of Corti, come in contact with bases of hair cells

5. **Types of hair cells**



Inner hair cells	Outer hair cells
one row of cells	3 rows of cells (more numerous)
Send 90 – 95 % of afferent auditory nerve	Send 5 – 10 % of afferent auditory nerve
	Receive mainly efferent cholinergic fibers from superior olive.
Primary sensory cells generate action potentials in auditory nerve → responsible for fine auditory discrimination.	Motile cells: shorten when depolarized & lengthen when hyper polarized No direct receptive function

➤ **Outer hair cells:** improve hearing by affecting vibration pattern of basilar membrane.

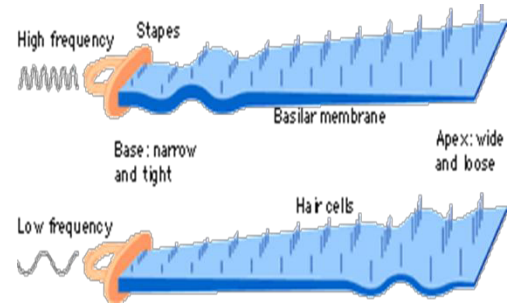
proof: High number of centrifugal fibers from brain stem to outer hair cells

→ suggest retrograde nervous mechanism for control of ear sensitivity to different sound pitches.

- Marked hearing loss occurs when outer hair cells are damaged, while inner hair cells remained fully functional.

Basilar Membrane

- ✓ Separates scala media from scala tympani,
- ✓ Consists of 20,000 to 30,000 **basilar fibers**
- ✓ Fibers:
 1. Stiff, elastic, reed like structures → Vibrate like reeds of a harmonica
 2. Basal end: Fixed to bony modiolus
 3. Distal ends: free [only attached by loose fibrous tissue]
 4. ↑ in length from base to top of cochlea
 5. ↓ in diameter from base to top of cochlea



➤ As a result:

Stiff short fibers near base of the cochlea (at oval window)	Long, lax fibers near apex (tip) of the cochlea	Fibers in between
Vibrate best at high frequency	Vibrate best at low frequency	Vibrate at frequencies in between

Presbycusis ↓ ability to hear high frequencies with age (may be due to loss of elasticity of short stiff basilar fibers)

Sound Transmission to organ of Corti via 2 routes

1. Ossicular route = Air route [ossicular conduction]

- **Main pathway** for normal hearing
- Sound waves → vibration of tympanic membrane → auditory ossicles → oval window (round window bulges out)
 - sound energy passes from scala vestibuli to scala tympani
 - vibrate basilar membrane → stimulating organ of Corti.

2. Bone route [Bone conduction]

- Via vibration of the skull by extremely loud sounds or vibrating objects (tuning forks)
 - causes vibration of the inner ear fluid

Mechanism of Stimulation of Auditory Nerve

1. **Propagation of sound wave to inner ear fluid:** via

2. **Movement of organ of Corti**

C- Sound waves spread from **oval window** (in & out motion) → along **scala vestibuli**

D- Most sound energy is transferred directly from scala vestibuli to scala tympani.

(Very little wave reaches apex)

→ cause vibration of basilar membrane & organ of Corti (up and down)

→ cause stereocilia (attached to tectorial membrane) to bend back and forth.

When organ of Corti moves up	When organ of Corti moves down
stereocilia bend away from limbus	stereocilia bend toward limbus

○ Tectorial and basilar membranes are attached at different points on modiolus (limbus)

→ slide past each other as they vibrate up and down)

3. **Generation of receptor potential:**

When stereocilia bend away from limbus	When stereocilia bend toward limbus
K ⁺ channels open → hair cell depolarizes.	K ⁺ channels close → hair cell hyperpolarizes

4. **Release of synaptic transmitter:**

Hair cell depolarization

→ opens voltage gated Ca²⁺ channel

→ Ca²⁺ entry

→ release of excitatory glutamate or aspartate

→ stimulates auditory nerve fiber.

5. **Potassium recycling:**

from hair cells into stria vascularis

via intracellular pathway

for secretion in scala media

Auditory Pathway

- Signals from both ears are transmitted on both sides, mainly opposite side.
- Collaterals pass into **RAS** → activates nervous system in response to loud sound.
- spatial orientation is maintained in fibers all the way from cochlea to cortex

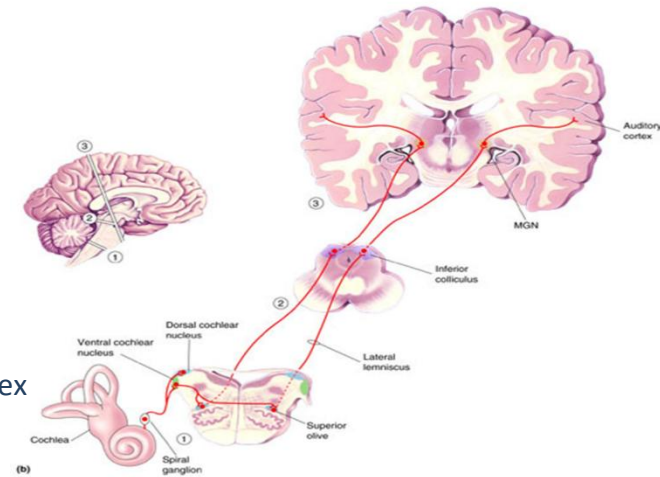
Auditory cortex

A- Primary auditory cortex

- ✓ Excited directly from medial geniculate body.

B- Auditory association cortex [2nd auditory cortex]

- ✓ Excited secondarily by impulses from 1ry auditory cortex



Auditory cortex is concerned with

- Perception of sound & psychic sensation of sound pitch
- Detect the direction of sound.
- Recognition of **sequential** pattern of sound. (during speaking or music listening)
- Centrifugal fibers from superior olivary nucleus mainly & auditory cortex to organ of Corti are inhibitory
→ allow direct attention to specific sounds &
rejecting other sound qualities
(listening to a single instrument within symphony orchestra)

Bilateral destruction of auditory cortex	Unilateral destruction of auditory cortex	Lesions in auditory association areas
Greatly ↓ sensitivity for hearing, Not cause complete deafness. Inability to recognize tonal pattern of sound & localization	Slightly ↓ hearing in opposite ear (NO deafness) because of many crossovers connection in auditory pathway	Inability to understand (interpret) meaning of sound (auditory aphasia)

Auditory Encoding

Definition

Ability of the auditory system to **discriminate** Sound

- **Frequency**
- **intensity**
- **localization**
- **Tonal & sequential** pattern of sound

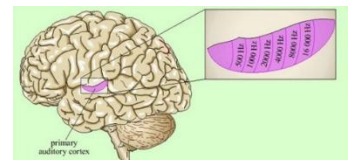
A. Determination of sound frequency ((pitch)

1. Place principle

- Sound wave produces peak depression & max. receptor stimulation at specific place of basilar membrane (its natural resonant frequency = sound frequency)
- Distance between this point & stapes is **inversely related** to frequency of sound

Low frequency	High frequency	Intermediate frequencies
Maximal stimulation at apex	Maximal stimulation at base	at intermediate distances

- Hair cells at different places are tuned to different frequencies because of differences in their **stereocilia & bio physical properties**.



2. Tonotopic Organization

- Spatial organization (orderly arrangement) of neurons in auditory pathway from cochlea to cerebral cortex

Neurons responding to low frequency sounds	Neurons responding to high frequency
located at one end of nucleus in the auditory pathway	located at opposite end

- Auditory cortex maintains topographic localization of sound

Low frequency sound	High frequency sound
Anteriorly	Posteriorly

B. Determination of sound Intensity (loudness): ↑intensity

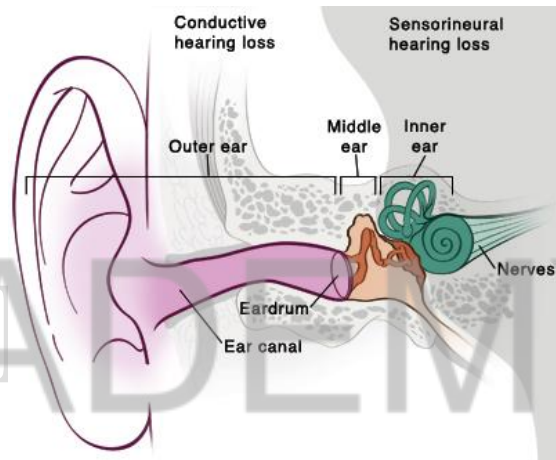
- 1) ↑ frequency of firing in the auditory nerve
- 2) Vibrate large portion of basilar membrane → more auditory nerve fibers are activated
- 3) Stimulate outer hair cells (when basilar membrane vibration reaches high intensity)
 - appraises nervous system that the sound is loud.

C. Determination of sound localization: depends on

- 1- **Intensity difference:** slightly more intense sound in the ear near to the source of sound than remote ear
- 2- **Phase (time) difference** in sound waves reaching 2 ears (At low frequencies)
- 3- **Pinnae** of 2 ears help in discriminating source of sound

D. Determination of sound patterns

- Auditory cortex recognizes combination or sequence of tones in specific pattern
(Respond specifically to shift from high to low frequency)
- **Lesions of auditory cortex: inability to recognize a patterned sequence of sound [as in music]**
(not impair ability to discriminate frequency)



Smell and taste

- Visceral senses (as being closely associated with GIT function).
- Related to each other as food Flavors depends on taste and smell
Depressed sense of smell in common cold → produce different food taste
- Both
 - ✓ Chemoreceptors, stimulated by molecules (dissolved in nasal mucus or saliva)
 - ✓ Have a unique ability to trigger **long-term memories**.

THE SENSE OF SMELL (OLFACTION)

Importance of Olfaction

1. Determines **Flavors** of food [together with taste, tactile, and thermal sense]
2. Essential for
 - **GIT functions** (reflexes)
 - **Avoiding dangerous** liquids or volatile substances.
 - **Sexual behaviour.**

sense of smell is not well developed in human compared to lower animals.

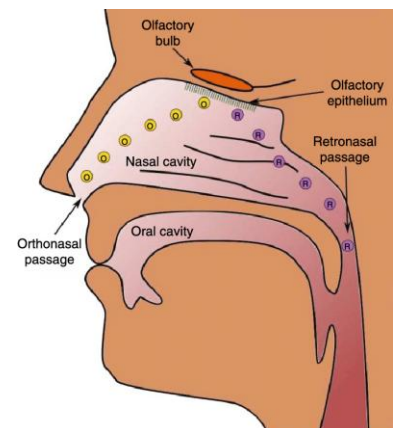
Olfactory Receptors

1. Location

- ✓ in olfactory mucosa (5 cm²)
Specialized nasal mucosa on superior nasal concha, adjacent to the nasal septum
- Not directly exposed to the flow of inspired air
- by sniffing → produces turbulence in airflow → Odorant molecules come in contact with olfactory mucosa

2. Innervation

- Olfactory nerve (cranial nerve 1)
- Free nerve ending of trigeminal nerve (cranial nerve V),
→ sense irritative character of some odorants

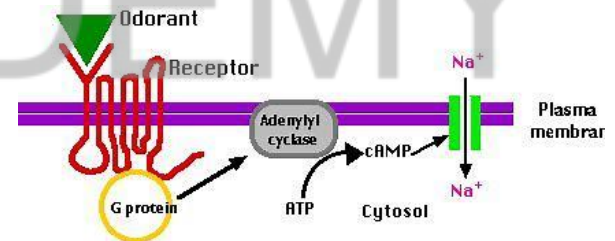


Properties of Olfactory receptors

1. **Sensitivity:** Extremely high, excited by Few molecules
2. **Discriminate 10,000 types of Odors**
3. **Discriminate intensity with limited ability** (odorant concentration in ambient air):
At least 30% change in concentration must occur before detection of the change.
4. **Adaptation:** very rapid (may be due to receptor adaptation or Central effect)
5. **Receptor potential** caused by adsorption of odorant molecules to the plasma membrane of the cilia of the receptor cells
6. **Pattern of stimulation**
A specific olfactory receptor responds to many Odors with variable responses
 - Sensory perception is based on the pattern of receptors stimulation
7. **Olfactory stimuli** (Odorant molecules): To be effective, must be
 - Volatile (easily vaporized)
 - Water-soluble to dissolve in nasal mucus

Stimulation of the olfactory cells

- Odorant substance diffuses into the mucus
- binds with receptor = Odorant-Binding protein (OBP)
 - activate G-protein
 - activate adenylyl cyclase or phospholipase C
 - opens many Na channels
 - Na entry
 - generate receptor potential



Smell Abnormalities

Anosmia	Hyposmia	Dysosmia
Absence of the sense of smell	↓ olfactory sensitivity	Distorted sense of smell

THE SENSE OF TASTE

Importance to select food according to desire and metabolic needs

Taste must be distinguished from flavor, which includes olfactory, tactile, and thermal, taste.

Types of taste sensations= 5 basic types

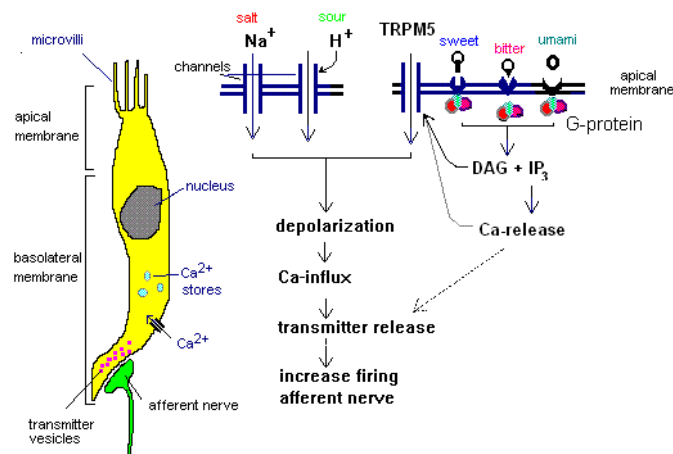
all taste sensations result from various combinations of these 4 primaries

	sweet sensation	bitter sensation	Salty sensation	Sour sensation	Umami
Produced by	Organic molecules as sugars, glycols, and aldehydes	Alkaloids as quinine and caffeine.	Ionizable salts	Acids	Protein rich Flavor ,produced by amino acids esp. glutamine Monosodium G
The area most sensitive	Tip of the tongue	Back of the tongue	Front half of each side of the tongue	Posterior half of each side of the tongue	Evenly distributed in the tongue

Taste buds Contain Taste Receptors.

- A- Distribution: on the tongue papillae, hard and soft palate, epiglottis, and pharynx
- B- Structure: Histology
- C- Specialization:

At low concentration	At high concentration
Each taste bud usually responds to only 1 of the 4 primary taste stimuli	Most buds can be excited by 2 or more of the primary taste stimuli



Mechanism of stimulation of taste receptors:

- ✓ Only chemicals in solution- either liquids or dissolved solids in saliva- can attach to receptor cells and evoke sensation of taste:

1-Sweet-tasting substances depolarize taste cells by:

- Opening of Na^+ channels
- Activating adenylate cyclase $\rightarrow$ CAMP $\rightarrow$ closure of K selective channels.

2- Bitter-tasting substances depolarize taste cells by:

- Stimulate the production of IPs $\rightarrow$ $\uparrow$ intracellular Ca

3-Salty-tasting substances depolarize taste cells by:

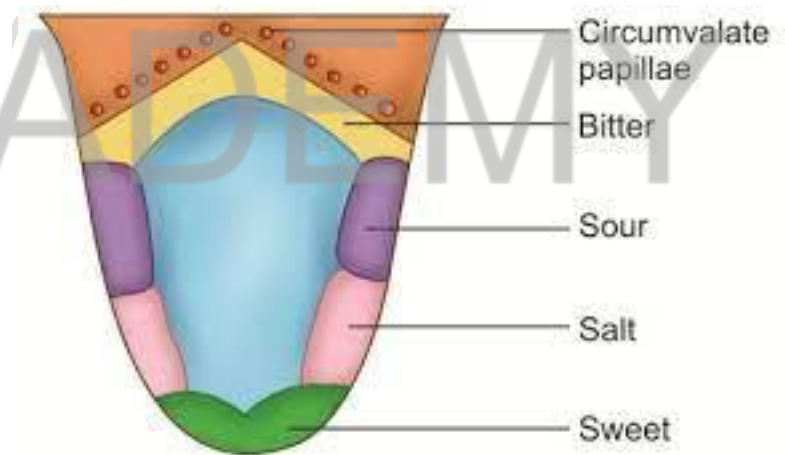
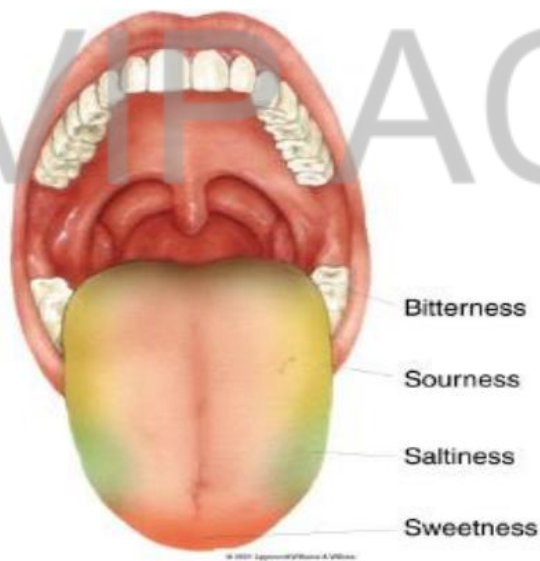
- Activating Na channels

4- Sour-tasting substances (e.g., citric acid) depolarize taste cells directly by

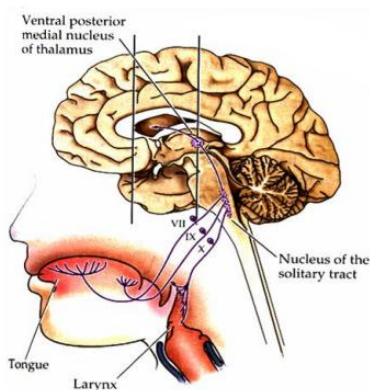
$\rightarrow$ Raising intracellular H^+ ion concentration, which blocks K^+ channels.

5-Umami (savory): monosodium glutamate binds to umami receptors $\rightarrow$ activate phospholipase C

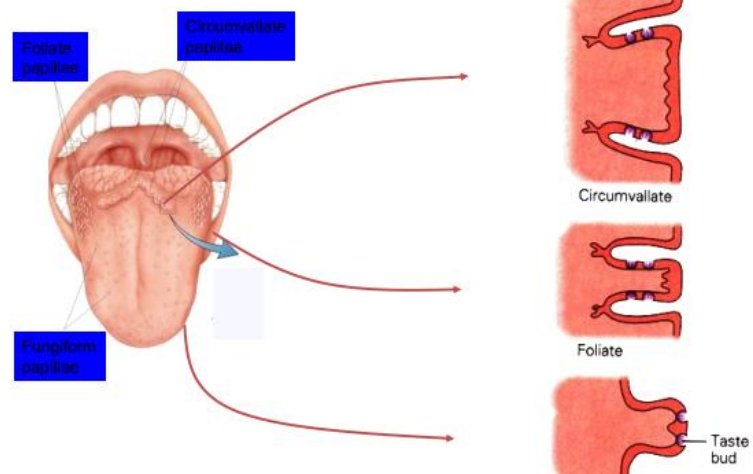
$\rightarrow$ $\uparrow$ intracellular Ca^{++}



From Tongue to Brain



Types of Papillae



Innervation of Tongue

